

Living Map

The Gafsa Mine Case

Tunnels: dangers, sensors, robots,
radio and the Command Post

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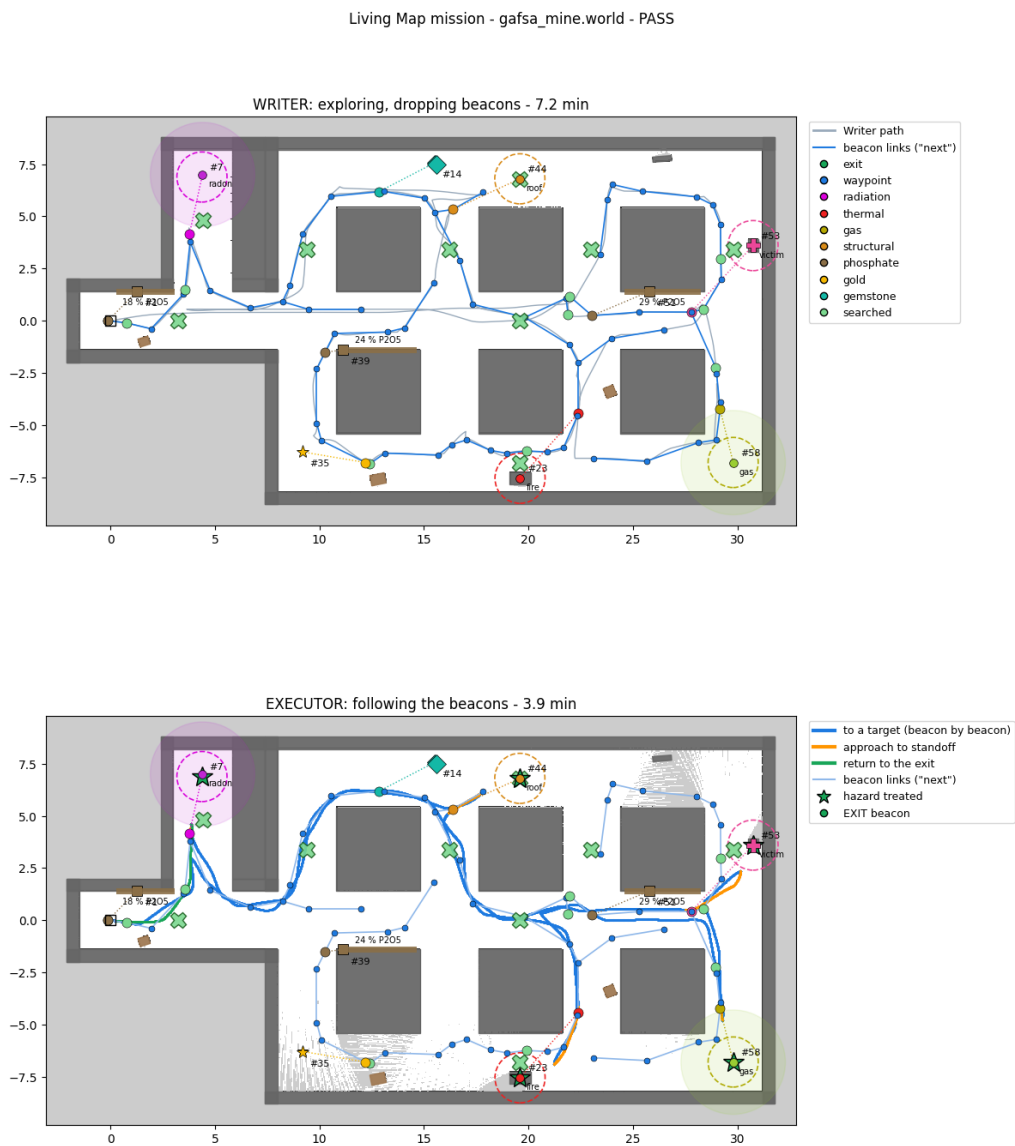
1. From a building to a mine: the Gafsa case

The Living Map was designed for a contaminated building. This document moves it **underground**, into the phosphate mines of the **Gafsa basin** in south-west Tunisia, and says what that changes: the dangers the robots meet, the sensors they need, the robots themselves, the radio, and what the Command Post shows. Everything described here is built into the delivery: a Gazebo mine, a mine sensor suite, new record kinds, a mine mode for both robots, the ONA and the dashboard, a scripted demo, and tests.

1.1 Why Gafsa, and which mine

- The Compagnie des Phosphates de Gafsa (CPG) mined underground from 1899 (Métlaoui), then Redeyef (1903), Moularès (1904) and M'dhilla (1920), with the **room-and-pillar** method ("chambres et piliers"). From the 1970s it moved to open pits; the last underground mine, Redeyef, closed around 2006 [1][2].
- About **70 old mining sites closed before 2007 were never rehabilitated** [3]. Old, unmaintained underground workings are therefore the realistic setting for a rescue robot: bad air, loose roofs, water, no light, no radio.
- USGS records the Métlaoui underground mine as room and pillar, 60 % recovery, a 4.5 m ore body, 5 shafts and one adit, at 34.3159 N, 8.4184 E [4]. The simulated panel is anchored there.

WATCH OUT The mine in the simulation is a **simulated panel**, not a CPG plan: an access drift from the portal, a dead-end gallery, and a room-and-pillar panel with six pillars (galleries 2.8 m wide, scaled to the robot). The real pillar sizes and gallery heights of the Gafsa mines are not published.



2. The dangers underground

Each danger below is in the simulated mine, measured by an instrument of the mine sensor suite, recorded by a beacon, shown on the dashboard with its measured value, and handled by the Executor.

danger	why in Gafsa	sensed by	alarm level	record / Executor
Unstable roof, loose slabs, spalling pillars	room-and-pillar workings in marls and clays; surface subsidence over the Moularès workings reached 7 cm in a year (1992-93) [5]; old timber decays	roof scanner (upward depth camera or tilted LiDAR) + camera	visible sag, cracks, fallen slabs	STRUCTURAL ("unstable roof"), sag in mm; never driven under; the Executor sets a roof prop

danger	why in Gafsa	sensed by	alarm level	record / Executor
Radon and gamma	the phosphate carries uranium (Métlaoui 57, Moularès 62 ppm) and radium-226 (285-580 Bq/kg) [6][7]; radon builds up in unventilated dead ends (Abu Tartor, Egypt: every reading above 1000 Bq/m ³ underground) [8]	continuous radon monitor + gamma probe	1000 Bq/m ³ (IAEA workplace); 300 Bq/m ³ logged (EU reference) [9]	RADIATION ("radon"), Bq/m ³ ; the Executor ventilates (fan + duct)
Bad air	blasting fumes (CO, NO, NO ₂); oxygen as low as 9.3 % where old workings were opened (MSHA) [10]; H ₂ S and CO ₂ possible (sulphides, gypsum, stagnant water); methane not documented in Gafsa but cheap to watch	multi-gas detector: CO, NO ₂ , H ₂ S, CH ₄ (%LEL), O ₂	CO 100 ppm, NO ₂ 5 ppm, H ₂ S 10 ppm, CH ₄ 20 %LEL, O ₂ < 19.5 % (portable detector high alarms) [11]	GAS ("bad air"), × the alarm level; the Executor ventilates
Fire	diesel loaders, conveyor belts, cables	thermal camera (+ CO)	hot spot	THERMAL ("fire"), °C; the Executor extinguishes
Trapped miner	the reason for the mission	thermal camera: body heat 30-37 °C	any	VICTIM; reached first ; oxygen and first aid
Is anybody there?	an empty gallery must be declared empty, not just "nothing seen"	thermal camera coverage, area by area	area 85 % seen	SEARCHED: "searched, N victims" (section 3.3)

2.1 Also there, not simulated

danger	what to do on the real robot
flooded workings: drowning is the most common death in abandoned mines [12]	a downward depth / sonar probe for water depth; mark as OBSTRUCTION ("blocked gallery"); IP67, wading depth
vertical shafts and winzes	downward cliff sensors (ToF) on all four corners: stop before an edge
old explosives	camera recognition, keep-out; never touched
respirable silica dust (the ore holds quartz, opal-CT) [13]	a dust sensor (PM4); a warning for the rescuers
heat and humidity	temperature / humidity on the gas head; a warning for the rescuers

3. The sensors, and what each record says

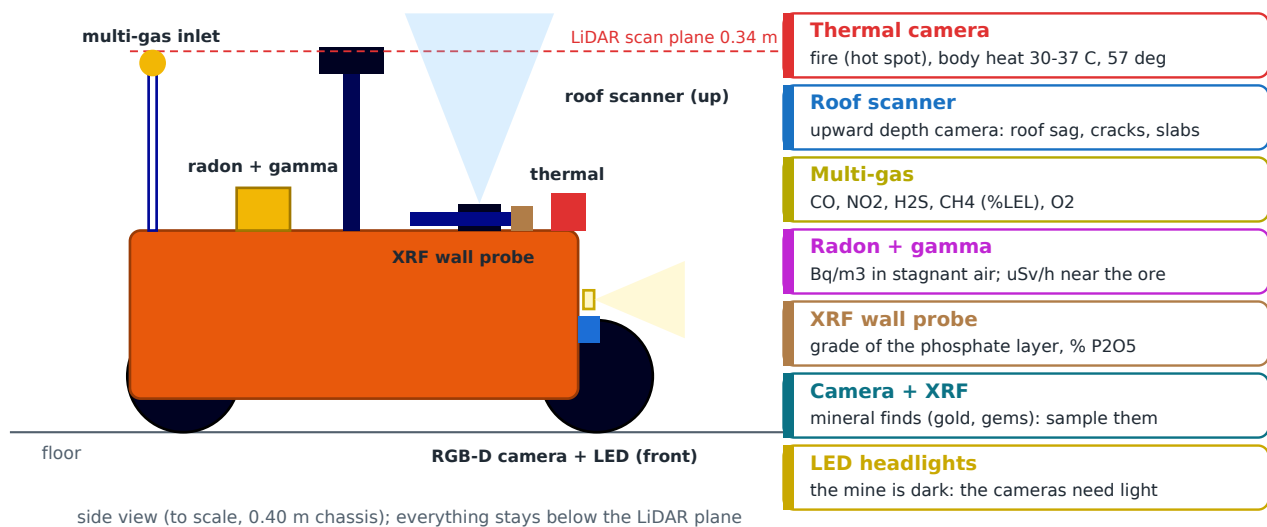


Figure 2. The mine sensor head (URDF option mine:=true, mounted automatically in the Gafsa world).

3.1 How the simulation measures

In the building, the camera recognises coloured blocks. Underground most dangers are invisible (gas, radon, a cracked roof) or in the dark (a fire, a person), so the mine has a **site file** (`worlds/gafsa_mine.json`) that lists them, and the `mine_sensors` node turns the robot's true pose into what each real instrument would report. It replaces the colour-block vision node in mine mode and publishes exactly what the vision node publishes (`/writer/hazard`, `/writer/events`), so the explorer, the beacon node and the Executor need no special case.

instrument	model in the simulation
multi-gas detector, radon monitor	concentration = peak × exp(−d / spread), d = the walking distance through the galleries (never through rock); 1 Hz on <code>/writer/air</code>
gamma probe	background 0.08 μSv/h, up to 0.22 μSv/h next to rich ore (grade-dependent)
thermal camera	57° view: a hot spot up to 8 m, a body up to 6 m, in line of sight
roof scanner	roof sag and cracks up to 3.5 m ahead, 120° view, in line of sight
wall probe (XRF / gamma)	a phosphate layer within 2 m: its grade, ± 0.6 % P ₂ O ₅
camera + XRF	mineral finds up to 3 m in the camera view, in line of sight
victim search	the thermal camera paints the floor it has seen (rays every 2°, stopped by rock); an area is searched at 85 %

A gas or radon source is never "seen": the robot localises it from the concentration gradient; the simulation gives the true source with a 0.25 m error, placed on the robot's own map from its own pose.

3.2 Record kinds and what the severity means underground

The 44-byte LMB2 record has one byte of severity. Underground each kind uses it for one physical value, so the Command Post can print the measurement, not a number:

kind (code)	severity means	half-life / lifetime	ONA uplink
GAS (2)	× the alarm level = severity / 20	10 min / 1 h	urgent
RADIATION (3)	radon ≈ severity × 50 Bq/m ³	6 h / 12 h	urgent
THERMAL (4)	hot spot ≈ severity × 6 °C	20 min / 1 h	urgent
STRUCTURAL (6)	roof sag ≈ severity × 0.6 mm	24 h / 3 d	normal

kind (code)	severity means	half-life / lifetime	ONA uplink
VICTIM (1)	condition (body heat seen)	1 h / 2 h	urgent
PHOSPHATE (8)	grade = severity / 4 % P ₂ O ₅ (0.25 % steps)	3 d / 7 d	low
GOLD (9), GEMSTONE (10)	a signal: sample to confirm	3 d / 7 d	low
SEARCHED (11)	% of the area seen by the thermal camera	2 h / 12 h	normal

Kinds 8 to 11 are new in the protocol (C core, firmware, Python codecs, ONA, robots, bptool, Windows .exe), all cross-checked byte for byte. Resources and searched areas are **marked, never avoided**: no keep-out zone, no retreat; the ONA sends resources last and never over the satellite link.

3.3 "Is there a victim or not": the search, area by area

Rescue teams spray an "X" at the door of every room they searched, with the number of victims found. The Writer does the same with a beacon. The site plan cuts the panel into **nine search areas** (the access drift, the dead-end gallery, the north, middle and south galleries, four groups of cross-cuts). The thermal camera paints what it has seen; when an area is 85 % seen the Writer drops a **SEARCHED** beacon: the record's position is the area's centre, its severity the coverage. The dashboard colours each area: **green** searched and nobody, **pink** a victim found there, **grey** not searched yet.

3.4 Resources: phosphate, gold, gemstones

- **Phosphate** is what Gafsa is: the Chouabine formation holds up to ten phosphate layers; the basin's usual classes are poor 15-22 %, medium 22-27 %, rich above 27 % P₂O₅; CPG sells 65-68 % BPL, about 30-31 % P₂O₅ [13][14]. The wall probe grades every layer the Writer passes; the simulated panel has a rich (29 %, layer C II), a medium (24.5 %) and a poor (18 %) layer. One record per 4 m of layer.
- **Uranium and rare earths** come with the phosphate (U 20-60 mg/kg, rare earths 114-600 mg/kg) [13][15]: the gamma probe already sees them; they are a by-product, not a separate kind.
- **Gold and gemstones: none is documented in the Gafsa basin** (the USGS lists no gold or gem production in Tunisia; the celestine of Tunisia is in the north) [16]. The panel has one of each as a **demonstration target** for the detector, labelled as such everywhere; in another mine the same code would mark real finds.
- The Executor treats nothing it was not asked to: resources become targets only when the Command Post briefs them ("taking a phosphate sample").

4. The robots for a mine

need	specification	in the delivery
explosive atmosphere	ATEX 2014/34/EU Group I, category M2 at least (M1 if methane is expected), IECEx Mb; flameproof enclosures (Ex d) or pressurised housings, intrinsically safe sensors (Ex ia); de-energise on a methane alarm [17]. Gafsa is not documented as gassy; old workings are unknown.	design requirement (hardware)
dust and water	IP67, sealed or nitrogen-pressurised electronics (like Numbat); wading depth for flooded floors (a rescue robot stopped at water in Pike River) [18]	requirement
ground	tracks or bigger wheels, ground clearance for rubble, low profile; tilt sensor	requirement
darkness	LED headlights + IR illuminator	URDF: headlights
perception	3D LiDAR (roof and walls), RGB-D, thermal camera, upward roof scanner	URDF + mine_sensors

need	specification	in the delivery
air	multi-gas: CO 0-500 ppm, NO ₂ 0-20 ppm, H ₂ S 0-100 ppm, CH ₄ 0-100 %LEL (infrared), O ₂ 0-25 %; radon monitor; gamma dose-rate probe; dust, temperature, humidity	URDF + mine_sensors
resources	XRF analyser on a wall probe, gamma spectrometer, UV / multispectral camera	URDF + mine_sensors
navigation	SLAM in repetitive galleries (identical pillars): the beacons are anchors; wheel + IMU EKF on the real robot	same SLAM; beacons
beacons	IP67, intrinsically safe, clip or magnet mount on a pillar, a beacon at every junction and corner	trail rule: corners + every 2.5 m
power	sealed LiFePO ₄ battery, enough for the way in and out	requirement

4.1 Radio underground

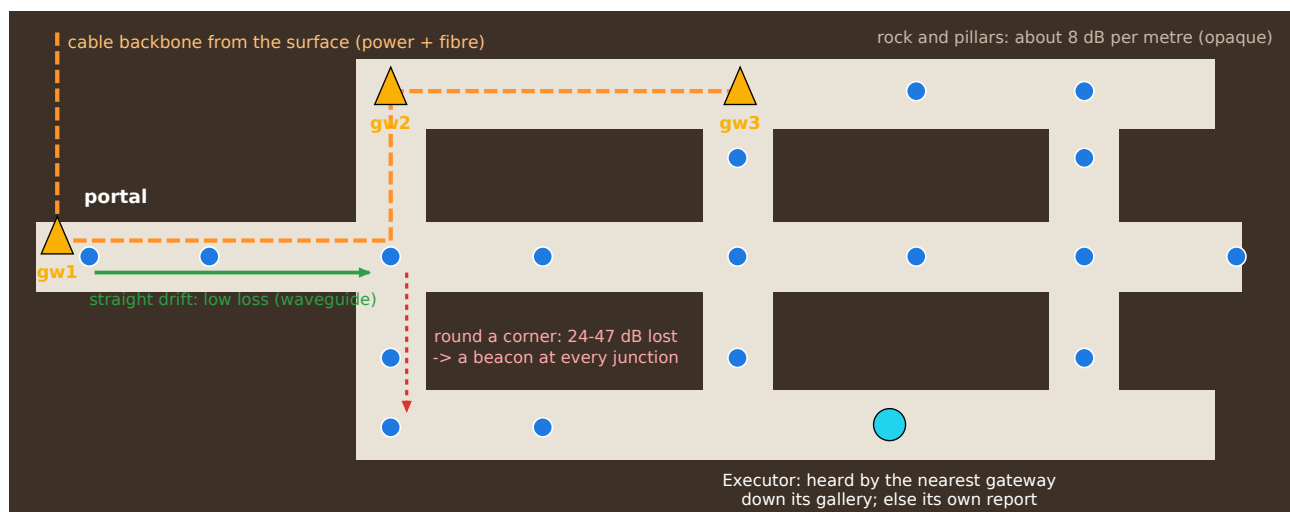


Figure 3. Radio in the panel: the gateways at the portal and on the mine's cable backbone; beacons at every junction.

- A straight gallery guides radio: LoRa at 868 MHz crossed more than 1000 m without line of sight in a potash room-and-pillar mine with under 15 % loss; but a 90° corner costs 24-47 dB [19][20]. Hence a beacon at every junction and corner (the Writer's trail rule already drops one at every turn).
- Rock is opaque: the simulation charges about **8 dB per metre** of pillar crossed.
- The three ONA gateways cannot stand "around the building": gw1 is at the portal, gw2 and gw3 on the mine's cable backbone at two junctions (powered and connected from the surface). They still vote 2-of-3 on every record: the beacons bring every record to every gateway.
- A gateway measures its distance to a robot (time of flight) **only straight down its gallery**: underground the ONA ignores signal-strength ranges. Where no gateway sees the robot, the dashboard shows the robot's own report ("no gateway in line of sight underground"). UWB ranging in mine drifts gives errors mostly under 1 m [21].

5. What the Command Post shows

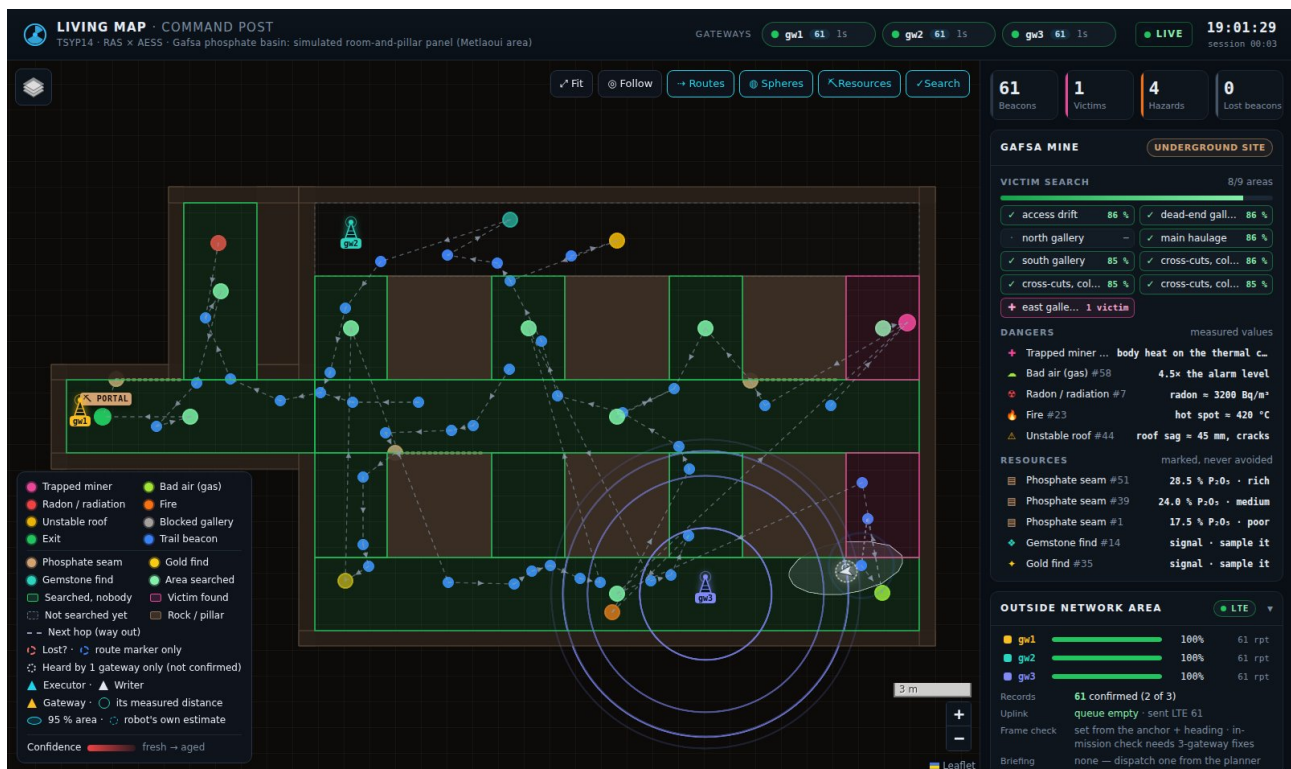


Figure 4. The Gafsa mine on the dashboard: the mine plan (rock, pillars, search areas), the beacons, the gateways underground, the Gafsa mine panel.

- **The mine plan is the map:** rock and pillars, the old plan's phosphate layers (dotted), the search areas, the portal; a dark rock background instead of street tiles (the surface is still one click away).
- **Mine words:** trapped miner, bad air (gas), radon / radiation, fire, unstable roof, blocked gallery.
- **Measured values** decoded from each record: "4.5× the alarm level", "radon ≈ 3200 Bq/m³", "hot spot ≈ 420 °C", "roof sag ≈ 45 mm", "28.5 % P₂O₅ · rich".
- **Gafsa mine panel:** the victim search (areas searched / victims found, click an area to zoom), the dangers with their values, the resources with their grades.
- **Resources** and **Search** buttons show or hide the resource pins and the area colours; new feed events "Resource marked", "Area searched".
- The ONA sends the site in its status (or `SITE=gafsa_mine npm start` without the ONA); the building scenario is unchanged.

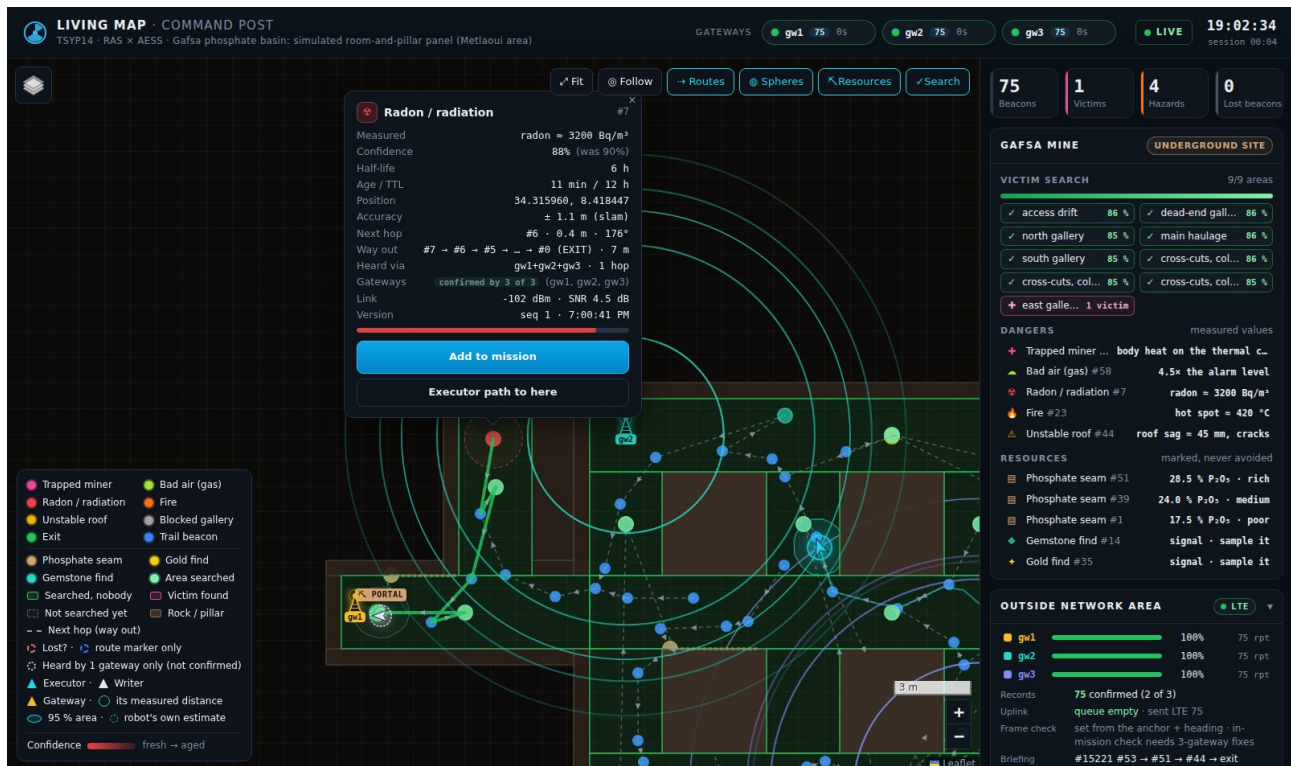


Figure 5. A record in the mine: radon measured at $\approx 3200 \text{ Bq/m}^3$, confirmed by the three gateways; the Executor (centre) on its briefing #53 $\rightarrow$ #51 $\rightarrow$ #44.

6. Run it

Install as in the Guide (parts A and B). Every command is also in `COMMANDS.txt`, part N.

6.1 Without ROS

Windows

`C:\tspy\living_map\beacon_net\windows\run_dashboard_mine_demo.bat`

the dashboard + the mine mission replayed: the plan, beacons appearing, areas turning green, then the Executor waiting at the portal: **Select on map $\rightarrow$ Dispatch mission** within a minute, else it goes for the miner first

Windows

`C:\tspy\living_map\writer_robot\windows\sim_mission_mine.bat`

the whole mine mission in the 2D simulator, with its checks and a picture

Windows

`C:\tspy\living_map\ona\windows\run_ona_mine.bat`

the real ONA for the mine (window 1; dashboard started first)

Windows

`C:\tspy\living_map\ona\windows\simulate_mine.bat`

window 2: the mine mission as the three gateways hear it

VM terminal

`cd ~/living_map/command_post && npm run demo:mine`
the dashboard mine demo in the VM

6.2 With the Gazebo robots

VM terminal

`~/writer_robot_ws/run_explorer.sh mine headless ona`

the Writer in the Gafsa mine: mine sensor head, mine sensors instead of the colour vision, beacons for dangers, resources and searched areas; the dashboard (on Windows) draws the mine plan

VM terminal

`tail -f /tmp/mine_sensors.log`
every 10 s: CO, NO₂, H₂S, CH₄, O₂, radon, gamma, sightings, victim search per area

VM terminal `ros2 topic echo /writer/air`
the air readings, once a second

VM terminal `~/writer_robot_ws/run_executor.sh headless ona`
after the Writer: the Executor in the same mine (the mission file says which world), victims first

VM terminal `~/writer_robot_ws/run_executor.sh mine headless ona`
the ready-made mine mission, no Writer run needed

6.3 Change the mine

Edit the layout, the dangers, the resources, the search areas or the gateways in `writer_robot/tools/make_mine.py`, then regenerate everything (world, site file, dashboard plan, ONA configs) and re-install:

VM terminal `cd ~/living_map/writer_robot && python3 tools/make_mine.py && bash install.sh`
rewrites the Gazebo world, `worlds/gafsa_mine.json`,
`command_post/public/sites/gafsa_mine.geojson` and the two mine ONA configs

7. Verification

test	what it checks in the mine	result
<code>tools/run_tests.py</code>	3 Writer runs in the mine (+ 18 in the buildings): every danger spotted, every resource graded, every area searched, no collision, never within 0.9 m of a danger, home	21 / 21
<code>tools/run_mission_test.py</code>	2 whole mine missions + the ready-made mission with a briefing (sample the rich layer, then the miner) and with $\pm 15\%$ sensor range errors (+ 21 building missions)	26 / 26
<code>tools/test_nodes.py</code>	the real mine_sensors node: air at the portal, radon in the dead end, a searched area, the rich layer graded, the miner, the fire, one event per danger, PHOSPHATE and SEARCHED records	70 / 70
<code>beacon_net: make test</code>	kinds 8-11 identical in C, Python, firmware	all pass
ONA tests	unchanged behaviour with the new kinds	70 + 20 + 3

A whole mine mission in 2D: the Writer explores the panel in about 7 simulated minutes (about 185 m), drops 75 beacons (5 dangers, 5 resources, 9 searched areas); the Executor reaches the trapped miner after 1.3 minutes, then treats the four other dangers and is back at the portal after about 4 minutes.

8. Limits, stated plainly

- The panel is simulated; real Gafsa gallery sizes, pillar layouts and ventilation are not public.
- The instruments are emulated from a site file: gas spreads along the galleries with a simple exponential model, with no airflow; a real mine has ventilation currents that move a plume one way.
- The rock radio model (8 dB/m, time of flight only in line of sight) is a first-order model; measure on site.
- Gold and gemstones are demonstration targets: none is documented in the Gafsa basin.
- Deep in a mine a robot can be out of every gateway's reach; the beacons would then have to relay its position reports and the briefings (ROBOT / MISSION frames). The protocol carries them; the beacon firmware does not relay them yet.
- Explosion protection (ATEX / IECEx), IP67 and wading are hardware certifications, not software.

Sources

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